
Stakeholder Deliverable

Client 3: Final Report

Stakeholder-facing summary of the refreshed PSID crisis-module package

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1 Executive Summary

This final report explains the refreshed PSID crisis-module package in stakeholder terms. The package now uses a simpler category model, a cleaner deployable questionnaire, and a more defensible explanation of why some items survive the final shortlist.

Headline takeaway

The current production run starts from 52 historical questions and produces a 28-question scored module, but the live deployable instrument contains 26 questions because the two selected demographic-only traceability rows are removed from routine field administration. The client-facing category model has been simplified to two labels only: **Generic** and **Specific**. Financial items now compete inside the broader **Specific** bank alongside disaster and pandemic items rather than appearing as a stand-alone final category.

2 Headline Results

Metric	Value	Calculation / rule	Why it is used and the assumption behind it
Total ranked questions	52	$\text{count}(\text{all rows}) = 52$	Starting universe for the refreshed package.
Selected scored questions	28	$\text{count}(\text{selected_for_module} = \text{True}) = 28$	Defines the scored module retained for auditability and review.
Deployable questions	26	$28 - 2 = 26$ after removing the selected demographic-only traceability rows	Defines the routine field instrument; assumes demographic context can be asked elsewhere or only when needed.
Selected scored minutes	29.98	$\sum \text{minutes across the 28 selected rows} = 29.98$	Confirms the analytic shortlist stays inside the 30-minute review budget.
Deployable minutes	28.82	$29.98 - 0.35 - 0.817 = 28.813 \approx 28.82$	Gives the paper-length upper bound for the deployable package.
Generic rows	3	$\text{count}(\text{ri_threshold_score} \geq 0.70) = 3$ after normalization	Keeps only the strongest portable questions in the universal core.
Specific rows	49	$52 - 3 = 49$	Preserves context-dependent detail inside one client-facing category.
Average R_i	2.201	$\text{mean}(R_i) = 2.201$	Benchmark statistic showing average signal efficiency.
Maximum R_i	5.667	$\text{max}(R_i) = 5.667$	Top efficiency value used in normalization.
Average P_i	2.561	$\text{mean}(P_i) = 2.561$	Benchmark for distinctiveness-adjusted priority.
Maximum P_i	7.860	$\text{max}(P_i) = 7.860$	Ceiling after rarity, construct, portability, and redundancy adjustments.

Table 1: Top-line performance metrics for the refreshed package.

3 Why the Count Changed From 28 to 26

Stage	Rows	Minutes	Calculation / rule	Why it matters
Scored module	28	29.98	Count and summed minutes for rows retained under the time budget	This is the full analytic shortlist preserved in the scored CSV and review documents.
Selected demographic traceability rows	2	1.17	SPC_AGE_YEARS (0.35) + SPC_DESCRIBE_SEX_GENDER (0.817)	These rows help with framing and optional routing, but they are weak direct crisis-signal items.
Deployable instrument	26	28.82	$28 - 2 = 26$ and $29.98 - 1.167 = 28.813 \approx 28.82$	The live field package asks only the highest-yield crisis items by default.

Table 2: The count changed at the deployment stage, not at the scoring stage.

What this means in practice

The package did not lose analytical coverage. The 28-row scored module is still the audited selection. The 26-row deployable version is simply the leaner field instrument that removes the two selected demographic-only questions from routine administration.

4 Figures and Visual Diagnostics

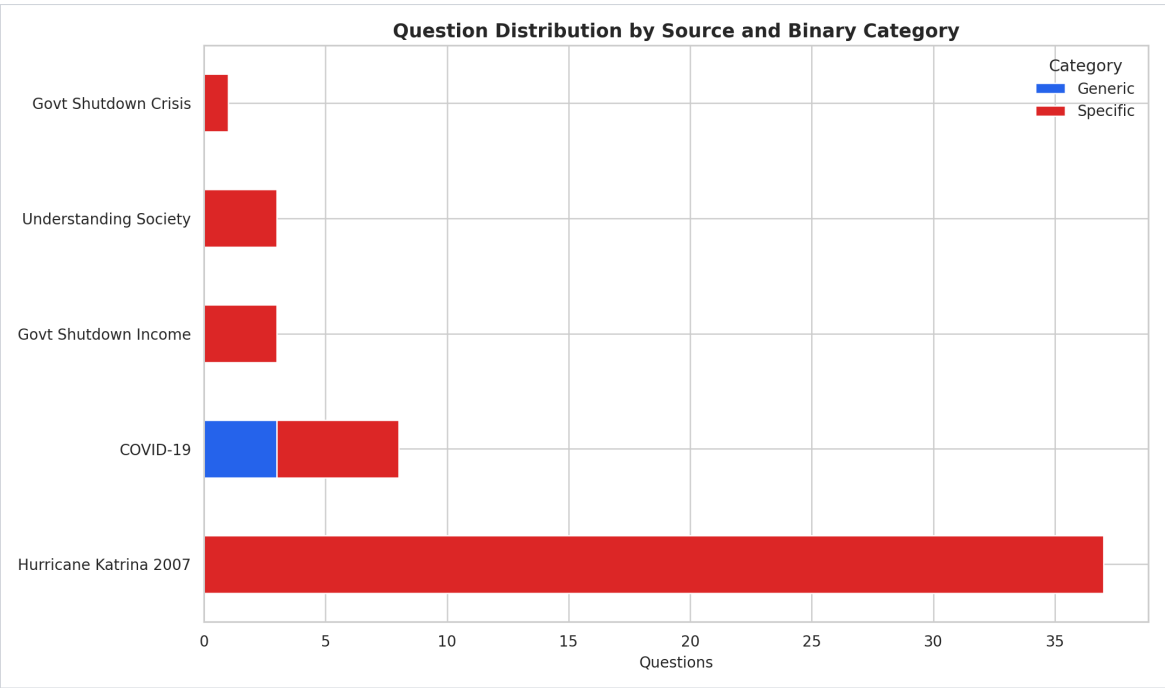


Figure 1: Binary Generic versus Specific distribution.

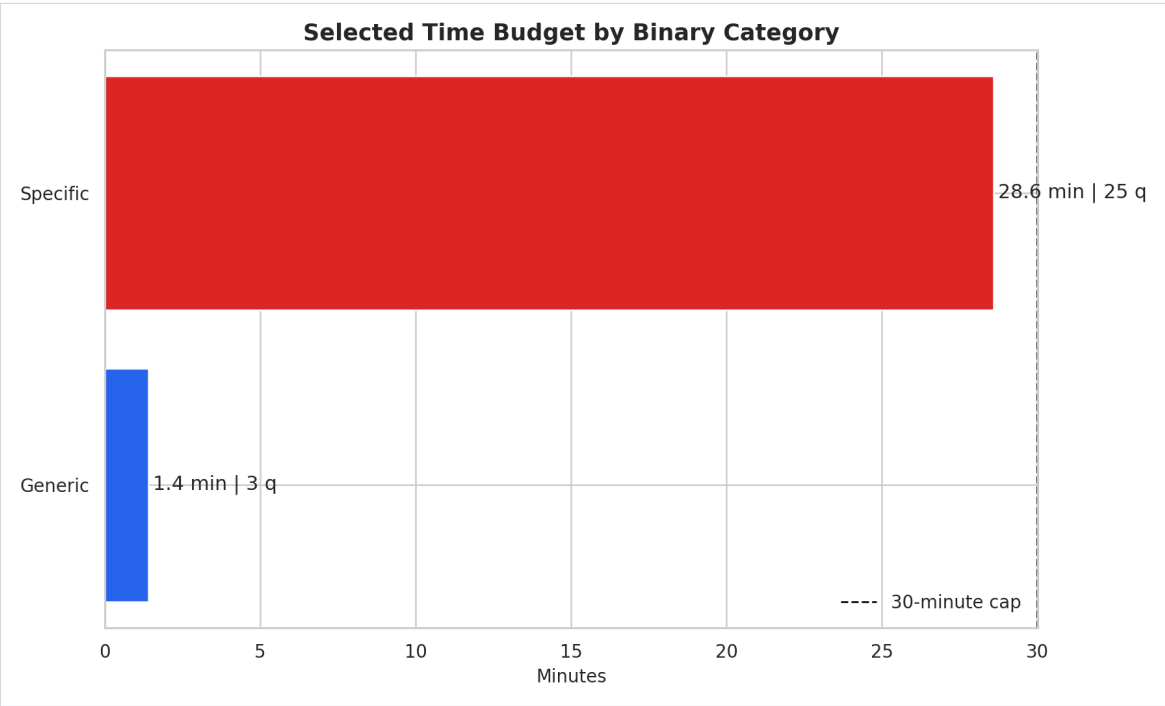


Figure 2: Time-budget allocation across the selected module.

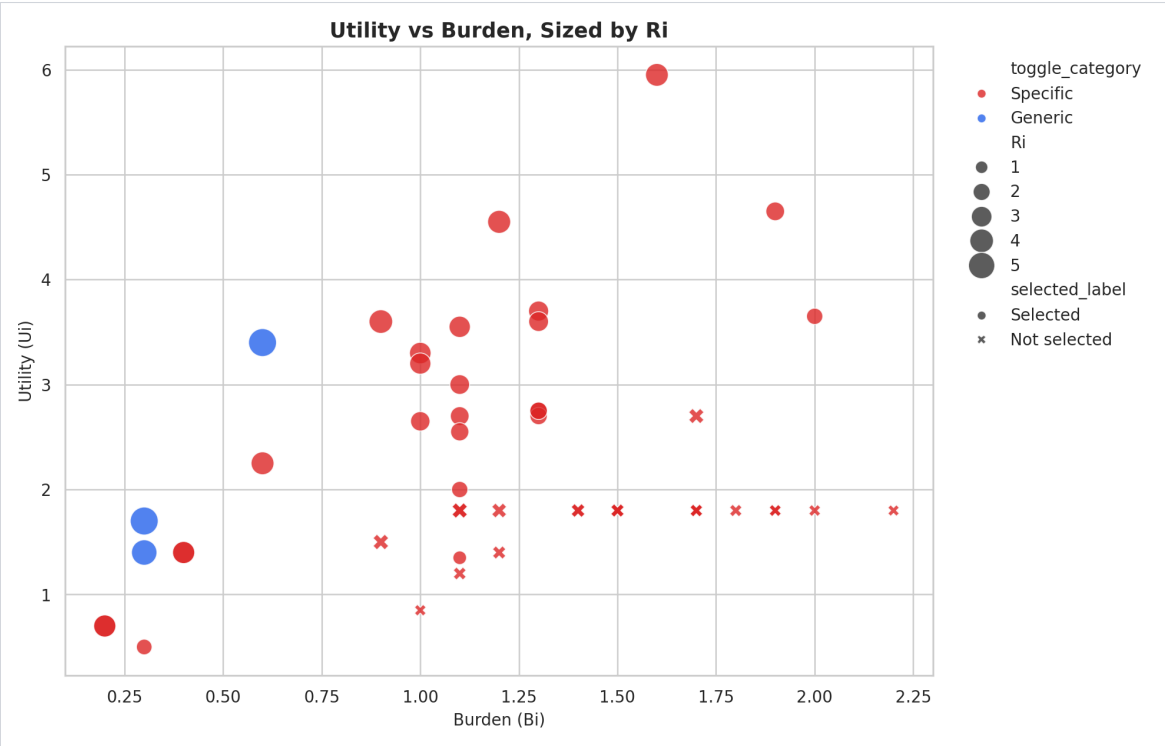


Figure 3: Utility-versus-burden relationship for the refreshed corpus.

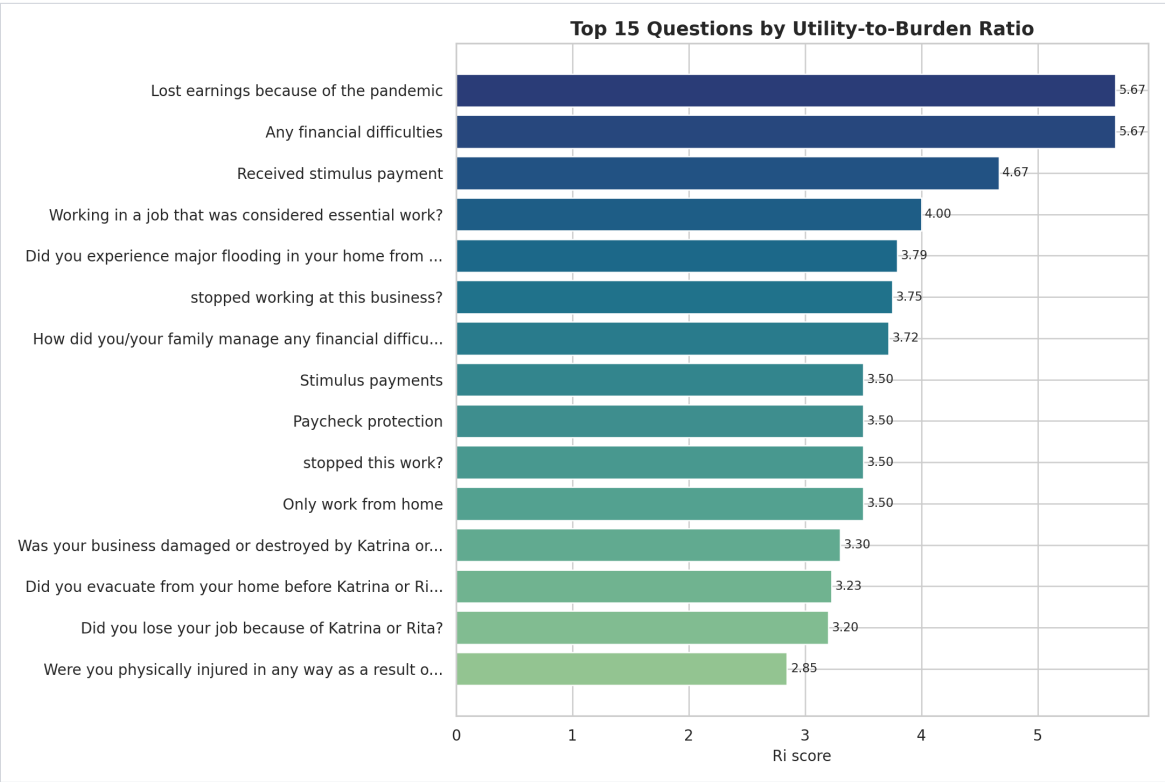


Figure 4: Top-ranked items retained in the final package.

5 Reporting Pipeline

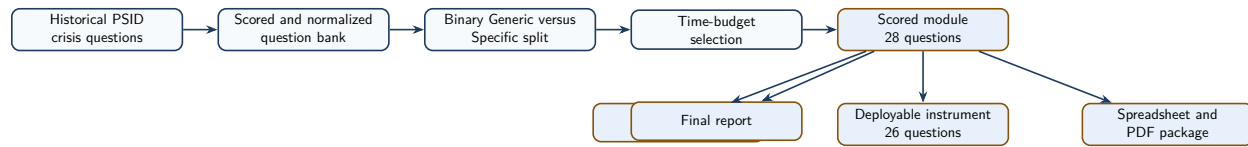


Figure 5: Reporting flow from historical question bank to the current client bundle.

6 What Changed in the Optimization Strategy

Dimension	Legacy architecture view	Current production view	Why it changed
Client-facing categories	Legacy architecture used 7 Generic Core rows, 1 Financial Crisis row, and 20 Pandemic / Disaster rows inside the older 28-row architecture	Current production uses 3 Generic and 25 Specific rows in the scored module, then 26 deployable rows after removing two demographic traceability items	The client package is easier to explain and the universal core is narrower and more portable.
Financial items	Finance appeared as its own visible label	Finance now competes inside Specific and is checked by redundancy penalties	Financial disruption remains covered without looking artificially dominant.
Deployment target	The scored shortlist was closer to the field-facing package	The scored shortlist is retained for traceability, while the field package is optimized separately for routine use	Separates analytic accountability from respondent burden.
Optimization levers	Ranking plus legacy toggle structure	Ranking plus portability bonus, construct coverage, redundancy control, documented wording replacements, and respondent-answer routing	The current package is optimized for a cleaner always-on instrument rather than for a broader archival architecture.

Table 3: Before-versus-now view of the optimization strategy.

7 Methodology in Plain Language

7.1 How ranking works

Each question receives a utility score for crisis relevance, a burden score for length and complexity, a baseline efficiency score R_i , and an augmented priority score P_i that rewards distinctiveness and construct coverage.

Main formulae

$$R_i = \frac{U_i}{B_i}$$

$$B_i = \max(0.10 \times \text{word_count} + 0.20 \times \text{complexity}, 0.1)$$

$$P_i = \frac{\text{augmented_utility}}{B_i \times (1 + 0.65 \times \text{redundancy_scaled})}$$

7.2 How the binary category is assigned

The client requested a simple threshold on R_i , but raw R_i already exceeds 0.70 for the entire refreshed corpus. The workflow therefore applies the cutoff to normalized R_i instead.

Operational thresholding

$$\text{ri_threshold_score} = \frac{R_i - \min(R_i)}{\max(R_i) - \min(R_i)}$$

$$\text{Generic} \iff \text{ri_threshold_score} \geq 0.70$$

$$\text{Specific} \iff \text{ri_threshold_score} < 0.70$$

7.3 How respondent answers now refine deployment

Refinement source	What changes quantitatively	Why it helps	Assumption
Generic-core answers and crisis identification	The effective interview path becomes the 3-item Generic core plus one crisis block; pandemic path is $1.40 + 2.92 = 4.32$ minutes, shutdown path is $1.40 + 4.20 = 5.60$, and disaster path is $1.40 + 20.30 = 21.70$	Respondents are not forced through irrelevant blocks, so burden falls without changing the underlying row scores	One dominant crisis context is active for a standard administration.
Selected demographic traceability rows	Removing age and sex/gender reduces the routine deployable length by 1.167 minutes	Keeps the instrument focused on direct crisis-impact content	Demographic context can be obtained elsewhere or only when needed.
Historical respondent distributions	Evidence on shutdown coping, stimulus use, and pandemic response justifies keeping aid, coping, and earnings-loss items salient	Grounds the optimization in observed behavior rather than only in wording similarity	Historical respondent patterns are informative for future crisis conditions.
Feed-forward respondent state	Prior-wave status, identity, address, and eligibility can refine follow-up universes without recalculating U_i , B_i , R_i , or P_i	Preserves longitudinal logic while keeping the ranking stable and auditable	Prior-wave metadata is available when the module is deployed longitudinally.

Table 4: Respondent-answer refinement changes the asked path, not the stored scoring fields.

8 Threshold Diagnostics

Diagnostic	Value	Calculation / implication
Minimum raw R_i	0.818	$\min(R_i)$ across the refreshed corpus.
Maximum raw R_i	5.667	$\max(R_i)$ across the refreshed corpus.
Rows with raw $R_i \geq 0.70$	52 of 52	A raw threshold would classify every row as Generic.
Rows with normalized threshold score ≥ 0.70	3 of 52	Normalization creates the intended narrow portable core.

Table 5: Why the Generic bucket is intentionally small after normalization.

This is why the current **Generic** bucket contains only the most portable high-signal items after normalization.

9 Selected Module Composition

Final category	Source	Selected questions	Minutes	Average R_i	Average P_i
Generic	COVID-19	3	1.40	5.333	7.125
Specific	Hurricane Katrina 2007	14	20.30	2.617	2.769
Specific	COVID-19	5	2.92	3.364	4.393
Specific	Govt Shutdown Income	3	2.33	3.326	4.482
Specific	Govt Shutdown Crisis	1	1.87	3.719	4.592
Specific	Understanding Society	2	1.17	1.447	2.329

Table 6: Composition of the current scored module by final category and historical source.

10 Why Financial Questions Do Not Dominate the Final Module

The earlier output looked more finance-heavy than the current one for two reasons. First, **Financial Crisis** used to appear as its own visible final label. Second, multiple finance-

related items were close variants of one another, which made the previous structure look broader than it really was.

Interpretation

- Financial items now compete within **Specific** instead of standing alone as a final category.
- Redundancy penalties reduce the advantage of repeating near-identical income questions.
- Katrina disaster items contribute many strong direct-exposure and trauma items.
- The three highest-portability Generic items still capture crisis-related financial disruption without making finance the whole story.

11 Demographic Note

Group	Rows	Average R_i	Average P_i
Demographics only	3	1.354	2.189
Non-demographic crisis items	49	2.253	2.584

Table 7: Demographic rows versus direct crisis-impact rows.

- Demographic rows provide context and traceability.
- They are not as efficient as direct crisis-impact questions for the deployable instrument.
- That is why the final scored module can retain them while the deployable questionnaire removes them.

12 Worked Example of a High-Value Generic Item

Example question: GEN_LOSE_EARNINGS_PANDEMIC

Field	Value	Calculation	Why it is used / assumption
U_i	3.400	$0.80 + 0.80 + 0.90 + 0.90$	Utility aggregates the matched crisis-signal weights; assumes the keyword tags capture the item's substantive value.
B_i	0.600	$\max(0.10 \times 6 + 0.20 \times 0.0, 0.1)$	Burden converts wording cost into a penalty term.
R_i	5.667	$3.40 / 0.60$	Very high baseline signal per unit of burden.
P_i	7.860	Value computed after rarity, construct, portability, and redundancy scaling	Strong augmented priority after distinctiveness adjustments.
minutes	0.70	$(6 \times 7) / 60$	Short enough to retain in the universal core; assumes 7 seconds per word in oral administration.
Final category	Generic	$\text{ri_threshold_score} = 1.000 \geq 0.70$	Portable across crisis types and strong enough to survive the binary threshold.

Table 8: Characteristics of a high-value Generic item.

This is the kind of item the new **Generic** bucket is meant to preserve: short, portable, and strongly informative.

13 Deliverables for Client Use

Deliverable	Use
docs/client 1.md	Technical Markdown codebook with formulas, figures, schema, and workflow notes.
docs/client 2.md	Markdown deployable questionnaire with live question tables and universe rules.
docs/client 3.md	Markdown final report for stakeholder circulation.
deliverables/Master_Questionnaire.xlsx	Full internal review sheet with traceability fields.
deliverables/Deployable_Questionnaire.pdf	Final deployable PDF questionnaire.
deliverables/Codebook.pdf	Rendered technical codebook PDF from the artifact pipeline.
deliverables/Final_Report.pdf	Rendered stakeholder-facing PDF from the artifact pipeline.
deliverables/Client_1_Technical_Codebook.pdf	LaTeX report version of the technical codebook.
deliverables/Client_2_Deployable_Questionnaire.pdf	LaTeX report version of the deployable questionnaire.
deliverables/Client_3_Final_Report.pdf	LaTeX report version of this stakeholder report.

Table 9: Client-facing outputs available after the LaTeX build.

14 Conclusion

The refreshed package is more disciplined than the earlier version in three ways:

- It keeps the category model easy to explain.
- It preserves the quantitative ranking logic instead of replacing it with ad hoc pruning.
- It separates the 28-row scored audit file from the 26-row deployable instrument, so burden reduction is explicit rather than hidden.

- It cleanly separates technical documentation, deployable wording, and stakeholder reporting.

The result is a client package that is easier to defend analytically and easier to use operationally.